Practical No. 7: Measures of Central Tendency and Dispersion for Grouped Data

Blue = input data from question; black = formulas. Midpoint x = (L+U)/2; h = class width; variance uses sample formula (n-1).

Ex.1 Mean, Median and Mode of age (years) of employees

Class	Lower L	Upper U	Freq f	Midpoint x	f·x	Cum. freq
20-30	20	30	7	25	175	7
30-40	30	40	16	35	560	23
40-50	40	50	15	45	675	38
50-60	50	60	9	55	495	47
60-70	60	70	3	65	195	50
Total			**50**		**2100**	

Result	Value
h (width)	10
Mean = Σfx/n	**42.00**
Median class	40-50
Median	**41.33**
Modal class	30-40
Mode	**39.00**

Formulas: Median = L + [(n/2 − cf)/f]·h ; Mode = L + [(f1 − f0)/(2f1 − f0 − f2)]·h. Expected: 42, 41.33, 39

Ex.2 Gasoline (gallons) sold per car, sample of 680 cars: Mean, Variance, Std. Deviation

Class	Lower	Upper	Freq f	Midpoint x	f·x	x²	f·x²	Result	Value
0-4	0	4	74	2	148	4	296	Mean = Σfx/n	**10.74**
5-9	5	9	192	7	1344	49	9408	Variance s²	**25.63**
10-14	10	14	280	12	3360	144	40320	Std. dev. s	**5.06**
15-19	15	19	105	17	1785	289	30345		
20-24	20	24	23	22	506	484	11132		
25-29	25	29	6	27	162	729	4374		
Total			**680**		**7305**		**95875**		

Formulas: Mean = Σfx/n ; $s^2 = [\Sigma fx^2 - n\cdot\bar{x}^2]/(n-1)$; $s = \sqrt{s^2}$. Expected: 10.74, 25.63, 5.06

Ex.3 Refrigerator lives (years): which model shows more uniformity? (lower C.V. = more uniform)

Life (yrs)	Midpoint x	f (A)	f·x (A)	f·x² (A)	f (B)	f·x (B)	f·x² (B)
0-2	1	5	5	5	2	2	2
2-4	3	16	48	144	7	21	63
4-6	5	13	65	325	12	60	300
6-8	7	7	49	343	19	133	931
8-10	9	5	45	405	9	81	729
10-12	11	4	44	484	1	11	121
Total		**50**	**256**	**1706**	**50**	**308**	**2146**

Measure	Model A	Model B
n	50	50
Mean	5.12	6.16
Variance s²	8.07	5.08
Std. dev. s	2.84	2.25
C.V. = s/mean × 100 (%)	**55.47**	**36.57**

Conclusion: Model B has the smaller C.V. (36.57% vs 55.47%), so it shows more uniformity.

Note: class midpoints are entered as inputs (blue). The answer key says Model B is more uniform.